

Lecture 1 - Dynamical Systems

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Time : $x: \mathbb{R} \rightarrow \mathbb{R}$, $t \in \mathbb{R} \mapsto x(t)$, $t \in \mathbb{R}$ is the time.

$t \rightarrow$ a continuous variable.

$x: \mathbb{Z} \rightarrow \mathbb{R}$, $k \in \mathbb{Z} \mapsto x_k$

$k \rightarrow$ discrete variable

I. CONTINUOUS DYNAMIC SYSTEMS are associated with DIFFERENTIAL EQUATIONS.

II. DISCRETE DYNAMIC SYSTEMS are associated with DIFFERENCE EQUATIONS.

I. CONTINUOUS D.S

1) Linear differential equations

↓
find solutions
+
analyse solutions

2) Dynamical systems associated to autonomous differential equations

- focus on dimensions 1, 2
- analyse solutions without a-priori finding them.

1) Linear differential equations

Examples: $x' = 0 \checkmark$
 $x' = -2t \checkmark$
 $x' = \sin t$
 $x'' = 0 \checkmark$

$x' - x = 0 \checkmark$
 $x'' - x = 0$
 $x'' + x = 0$

$\Rightarrow x$ is a function ($x(t)$)
"find functions whose derivative = 0"

$\bullet x' = 0$ / integrate $\Rightarrow \int x' dt = \int 0 dt = c$

$\Rightarrow x = c, c \in \mathbb{R}$ "general solution"

$\Rightarrow S = \{c \cdot 1, c \in \mathbb{R}\}$ is a subspace of $C(\mathbb{R})$
↓
the constant function 1

- the set of all solutions = lim. space of dimension = m (max order)
- the general form of LHDE
- IVP (=) bc. conditions = ODE order
- III! Theorem for LHDE:
IVP $\rightarrow$ unique sol.

$C(\mathbb{R})$ = the set of all continuous functions

Linear map: $f(k_1 v_1 + k_2 v_2) = k_1 \cdot f(v_1) + k_2 \cdot f(v_2)$

Subspace: $S \leq_K V$ $\begin{cases} S \neq \emptyset \\ \textcircled{a}) v_1, v_2 \in S, v_1 + v_2 \in S \\ \textcircled{b}) k \in K, v \in S, k \cdot v \in S. \end{cases} \rightarrow \begin{cases} S \neq \emptyset \\ \textcircled{a}) k_1, k_2 \in K, k_1 v_1 + k_2 v_2 \in S. \\ \textcircled{b}) v_1, v_2 \in S \end{cases}$

• $x' = -2t$ / integrate $\Rightarrow x = -t^2 + c$; $c \in \mathbb{R}$ "general solution".

$$\int x' dt = \int -2t dt = -2 \cdot \frac{t^2}{2} + C = -t^2 + c.$$

• $x'' = 0$ / integrate $\Rightarrow x' = c_1$ / integrate $\Rightarrow x = c_1 \cdot t + c_2$; $c_1, c_2 \in \mathbb{R}$

$$\int x'' dt = \int 0 dt = c_1$$

$$\int x' dt = \int c_1 dt = c_1 \cdot \int 1 dt = c_1 \cdot t + c_2$$

$S = \{c_1 \cdot t + c_2 \cdot 1; c_1, c_2 \in \mathbb{R}\} \rightarrow$ linear space of dimension 2.

Linear space = Vector space : K^V $\left\{ \begin{array}{l} (V, +) \text{ abelian group} \\ \text{ext. op. } \left\{ \begin{array}{l} 1) \dots \\ 2) \dots \\ 3) \dots \\ 4) \dots \end{array} \right. \end{array} \right.$

• $x' - x = 0 \Leftrightarrow x' = x$.

Find solutions ("euclidean liber") : $e^t \rightarrow$ function that verifies the eq.
 $0 \rightarrow$ another " — " — "
 $c \cdot e^t, c \in \mathbb{R} \rightarrow$ " — " — "

Obs. that : $1 \cdot e^t = e^t$
 $0 \cdot e^t = 0 \Rightarrow e^t, 0 \in \{c \cdot e^t, c \in \mathbb{R}\} = \mathbb{R}$

Aim: find all the solutions of the differential equation.

Are there other solutions? No.

Justification:

$\rightarrow$ IDEA 1: $x' - x = 0 \quad | \cdot e^{-t}$

$$\Rightarrow x' \cdot e^{-t} - x \cdot e^{-t} = 0$$

$$\Rightarrow (x \cdot e^{-t})' = 0 \quad | \int$$

$$\Rightarrow \int (x \cdot e^{-t})' dt = \int 0 dt$$

$$\Rightarrow x \cdot e^{-t} = c \quad | \cdot e^t$$

$$\Rightarrow x = c \cdot e^t, c \in \mathbb{R}, \text{ So there are no other solutions!}$$

$$\int (\dots)' dt = (\dots) + C \quad (\text{primitive})$$

IDEA 2: Let S be the set of all solutions of $\underline{x}' - x = 0$.

We know: $\Sigma \subseteq S$. (*)

- $x \in \mathcal{S} \Leftrightarrow \exists c \in \mathbb{R}$ s.t. $x = c \cdot e^t \mid \cdot e^{-t}$

$$(\Rightarrow) \exists c \in \mathbb{R} \text{ s.t. } x \cdot e^{-t} = c$$

- Start with $x \in S$, prove it is in this set ($\tilde{S}$).

$x \in S; y = x \cdot e^{-t} \mid' \Rightarrow y' = x' \cdot e^{-t} - x \cdot e^{-t} \Rightarrow y' = x \cdot e^{-t} - x \cdot e^{-t} = 0$
 $\downarrow$
 $x \in S \Rightarrow x' = x \Rightarrow$ integrate $\Rightarrow y = c,$
 $c \in \mathbb{R}.$
 new variable $\rightarrow$ show v is a constant function

new variable $\rightarrow$ show y is a constant function
(function)

$$\Rightarrow \underline{x \cdot e^{-t} = c = y \in \mathbb{R}} \Rightarrow x \in \mathbb{R}$$

So $S \subseteq Z \Rightarrow Z = S.$

From (*) $\Rightarrow \Sigma \subseteq S$

→ IDEA 3: S is a subspace of $C(\mathbb{R})$. I know that S is a linear space of dimension 1.

(Linear algebra) $\tilde{S} = \{c \cdot e^t, c \in \mathbb{R}\} \subseteq S.$

$$S^2 = \langle e^t \rangle \Rightarrow S^2 \text{ linear space of dim 1.}$$

(THEOREM).

↓ (span)

S is also $\frac{1}{2} \frac{1}{2} \frac{1}{2} \frac{1}{2}$.

↓
lin. space generated by one element.

We conclude that $\mathcal{Z}_S = S$.

$$\bullet \quad x'' - x = 0$$

$0, e^t, c.e^t, e^{-t}, c.e^{-t}, \cosh t, \sinh t \rightarrow$ are all solutions.

$$\rightarrow x = e^{-t} \quad \swarrow$$

$$\Rightarrow x' = -e^{-t} / 1$$

$$\Rightarrow x'' = -(-1) \cdot e^{-t} = e^{-t} = x$$

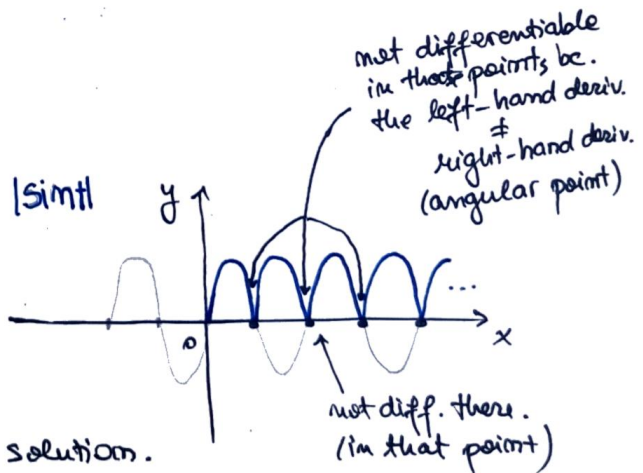
$$\Rightarrow x'' = x \text{ (so } e^{-t} \text{ is a solution).}$$

→ $x = |\sin t|$ solution? (no)

$$x = |\sin t| = \begin{cases} \sin t, & t \in I_1. \\ -\sin t, & t \in I_2. \end{cases}$$

1st branch $\rightarrow (\sin t)' = \cos t$

$\xrightarrow{\text{2nd branch}} (\sin t)'' = -\sin t \neq |\sin t| \Rightarrow \text{not a solution.}$



Q: Are cht and sh t solutions of $x'' - x = 0$? (yes)

(cosinus and sinus hyperbolic)

$$\begin{aligned} \text{cht} &= \frac{1}{2}e^t + \frac{1}{2}e^{-t} \\ \text{sh t} &= \frac{1}{2}e^t - \frac{1}{2}e^{-t} \end{aligned}$$

$$\begin{aligned} (\text{cht})' &= \text{sh t} \\ (\text{sh t})' &= \text{cht} \end{aligned}$$

$$\begin{aligned} \text{ch } 0 &= 1 \\ \text{sh } 0 &= 0 \end{aligned}$$

$$\text{ch}^2 t - \text{sh}^2 t = 1$$

To think: Why is it called "hyperbolic"?

Theorem: The set of all solutions of $x'' - x = 0$ is a linear space of dim 2.
(it has a base with 2 elements)

$\Rightarrow$ We need two linearly independent solutions of $x'' - x = 0$.

Take two (random) ones (to check): e^t, cht
(from the solutions)

$\rightarrow$ are they lin. ind.?

the zero function

Let $\alpha, \beta \in \mathbb{R}$ s.t. $\alpha \cdot e^t + \beta \cdot \text{cht} = 0, (\forall) t \in \mathbb{R}$.

We want $\alpha = \beta = 0$ $\Rightarrow$ lin. ind.

it is sufficient to prove this
" $\Rightarrow$ "

Choose $t=0 \Rightarrow \begin{cases} \alpha + \beta = 0 \\ \alpha \cdot e + \beta \cdot \text{ch } 1 = 0 \end{cases} \leftarrow \text{LINEAR HOMOGENEOUS SYSTEM.}$

$$\begin{aligned} \begin{vmatrix} 1 & 1 \\ e & \text{ch } 1 \end{vmatrix} &= \text{ch } 1 - e = \frac{1}{2}e + \frac{1}{2}e^{-1} - e = \frac{1}{2}(e + e^{-1}) - e = \\ &= \frac{1}{2}\left(\frac{1}{e}\right) = -\frac{1}{2}e + \frac{1}{2}e^{-1} = \frac{1}{2}\left(\frac{1}{e} - e\right) = \frac{1}{2} \cdot \frac{1 - e^2}{e} \neq 0 \\ & (e = 2, 7, \dots) \end{aligned}$$

$\Rightarrow \det \neq 0 \Rightarrow \alpha = \beta = 0 \Rightarrow$ the two functions (e^t, cht) are linearly independent.

($\beta \neq 0 \Rightarrow \text{cht} = -\frac{\alpha}{\beta} \cdot e^t$?! why)

Then, $x = c_1 \cdot e^t + c_2 \cdot \text{cht}; c_1, c_2 \in \mathbb{R}$.

We have that e^t and e^{-t} are lin. ind. solutions of $x'' - x = 0$.

also $\cosh t$ and $\sinh t$ " " " "

Thus, the "general solution" of $x'' - x = 0$ can also be described by:

$$x = c_1 \cdot e^t + c_2 \cdot e^{-t}; \quad c_1, c_2 \in \mathbb{R}$$

or

$$x = c_1 \cdot \cosh t + c_2 \cdot \sinh t; \quad c_1, c_2 \in \mathbb{R}$$

$$\bullet x'' + x = 0$$

Theorem: The set of all solutions is a linear space ^(vector space) of dim 2.

! Although it's a solution, 0 is linearly dependent w any other sol...

$\cos t$, $\sin t$ are solutions, lin. ind.

$\Rightarrow$ the "general solution" is:

$$x = c_1 \cdot \cos t + c_2 \cdot \sin t; \quad c_1, c_2 \in \mathbb{R}.$$

The general form of an m^{th} order Linear Homogeneous Differential Equation (LHDE)

$$(1): x^{(m)} + a_1(t) \cdot x^{(m-1)} + \dots + a_{m-1}(t) \cdot x' + a_m(t) \cdot x = 0$$

derivative of
order m^{th} w.r. to t

derivative of order 1^{st} w.r. to t

$\bullet m \in \mathbb{N}, m \geq 1 \rightarrow$ the ORDER of the differential equation (1).

$\bullet a_1, \dots, a_m \rightarrow$ they are called COEFFICIENTS

$\rightarrow$ continuous functions $\in C(I)$, $I \subset \mathbb{R}$
(open interval)

$\rightarrow$ when $a_1, \dots, a_m$ are constant functions, we say that (1) has "constant coeff".

! The Initial Value Problem (IVP) for (1):

Let $t_0 \in I$ fixed. Let $\eta_1, \dots, \eta_m \in \mathbb{R}$

$\downarrow$
initial time

$\downarrow$
"eta"

IVP $\Leftrightarrow$ nr. conditions = ODE order.

IVP $\left\{ \begin{array}{l} \text{eq. (1)} \end{array} \right.$

$$x(t_0) = \eta_1, x'(t_0) = \eta_2, \dots, x^{(m-1)}(t_0) = \eta_m$$

(initial conditions)

The existence and uniqueness theorem for LHDE:

The (IVP) has a unique solution, $\varphi: I \rightarrow \mathbb{R}$.

Def: Solution of eq. (1) is a function $\varphi: I \rightarrow \mathbb{R}$ s.t.

$$\begin{cases} \varphi \in C^m(I) \rightarrow \text{continuous + deriv. up till order } m \\ \varphi^{(m)}(t) + a_1(t) \cdot \varphi^{(m-1)}(t) + \dots + a_m(t) \cdot \varphi(t) = 0, (\forall) t \in I. \end{cases}$$

The Fundamental Theorem for LHDE

The set of all solutions of eq. (1) is a linear space of dim m . ^(vector space)

Proof:

$C(I)$ = the set of all cont. functions $\rightarrow$ has a structure of linear space.

$$C^n(I) = \{ x: I \rightarrow \mathbb{R} \text{ s.t. } \exists x, x', x'', \dots, x^{(m)}: I \rightarrow \mathbb{R}, \text{ cont.} \}$$

$C^m(I) \rightarrow$ subspace of $C(I)$.

$$x \in C^m(I); \quad L(x): I \rightarrow \mathbb{R}$$

$$\text{(function)} \quad ?! \quad L(x)(t) = x^{(m)}(t) + a_1(t) \cdot x^{(m-1)}(t) + \dots + a_m(t) \cdot x(t), \\ t \in I.$$

$$L: \underbrace{C^m(I)}_{\text{lin. space}} \rightarrow \underbrace{C(I)}_{\text{lin. space}}$$

Linear map: $L(\alpha x + \beta y) = \alpha L(x) + \beta L(y)$,
(char. theorem) $(\forall) x, y \in C^m(I)$
 $(\forall) \alpha, \beta \in \mathbb{R}$.

V vector space

$K = ?$ in this case?

$V = ?$ where are the scalars from?